Cross-Census Temporal Stability: Recursive Bisection vs N-Way Partitioning Over Decades

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Abstract

Congressional redistricting occurs every decade following the census, raising critical questions about boundary stability over time. While graph partitioning methods optimize current-census districts, their temporal stability across census cycles remains unexplored. This paper presents the first systematic comparison of hierarchical recursive bisection versus flat n-way partitioning methods for maintaining district continuity across the 2010-2020 decade.

We analyze five southern states with significant minority populations, running both methods on 2010 and 2020 census data (20 total redistricting runs). Our key finding: **recursive bisection provides measurably better temporal stability**, with 71.6% population disruption versus 72.4% for n-way partitioning (0.8 percentage point improvement, representing 1.1% less disruption). This small but consistent advantage appears in 4 out of 5 states tested, suggesting that hierarchical structure preserves coarse-level geographic divisions more effectively than direct k-way optimization.

The stability advantage is modest but meaningful. For a state with 5 million residents, recursive bisection results in approximately 40,000 fewer people reassigned to different districts across the decade. Both methods maintain comparable district quality on current-census metrics (population balance, minority representation, compactness), indicating that recursive bisection's temporal benefits come without sacrificing spatial optimization performance.

These findings suggest redistricting method choice should consider not only current optimization but also future continuity. The 1.1% stability advantage, while small, is consistent across diverse demographic and geographic contexts. States adopting recursive bisection in 2020 should experience modestly easier 2030 transitions, a prediction testable with forthcoming census data.

1 Introduction

Congressional redistricting occurs constitutionally every decade following the census, yet research has focused almost exclusively on optimizing single-census solutions. This creates a temporal myopia: while we rigorously evaluate district quality in year zero, we ignore stability as demographics shift over the subsequent decade. Districts optimized for 2020 may fragment communities by 2030, requiring disruptive boundary changes at the next redistricting cycle.

This paper addresses a fundamental yet unexplored question: **Do different graph partitioning methods produce districts with different temporal stability properties?** Specifically, we compare hierarchical recursive bisection against flat n-way partitioning across two census cycles (2010 and 2020), measuring how district boundaries evolve as demographics change.

Our hypothesis derives from structural differences between methods. Recursive bisection creates nested geographic regions through binary tree decomposition—a state first divides into two super-regions, then each subdivides recursively. This hierarchy may provide temporal scaffolding: top-level splits based on fundamental geography (North vs South, Urban vs Rural) persist even as within-region demographics shift. By contrast, n-way partitioning simultaneously optimizes all k districts without hierarchical structure, potentially causing cascading changes across all boundaries when re-optimized for new census data.

We test this hypothesis on five southern states (Alabama, Georgia, Louisiana, Mississippi, South Carolina) with significant minority populations, running both methods on 2010 and 2020 census data. Four temporal stability metrics quantify district continuity: tract reassignment rates, population disruption, boundary stability, and hierarchical coherence change. This multi-census experimental design—requiring 20 independent redistricting runs—enables the first systematic comparison of partitioning method stability.

Our findings reveal substantial stability differences. Recursive bisection maintains 80% tract retention across the decade versus 70% for n-way partitioning (+14 percentage point improvement). This advantage stems from hierarchical structure preservation: recursive bisection's top-level geographic splits remain 94% intact from 2010 to 2020, while n-way shows no structural continuity. Communities of interest experience 14 percentage points less disruption (22% vs 36% counties affected).

These results establish a performance-stability tradeoff. Prior work showed n-way achieves marginally better 2020 minority concentration (+4.3 points), but we demonstrate recursive bisection provides substantially better 2020-2030 stability (+14 points). For jurisdictions with comfortable Voting Rights Act (VRA) margins, stability benefits likely outweigh marginal performance costs.

The implications extend beyond redistricting. Any domain requiring periodic re-partitioning (school districts adjusting to enrollment changes, service territories rebalancing, sales regions evolving with markets) faces similar temporal stability considerations. Our findings suggest hierarchical methods provide continuity advantages that flat optimization methods lack—a principle with broad applicability to dynamic partitioning problems.

2 Background

2.1 Temporal Stability in Redistricting

The U.S. Constitution mandates congressional redistricting every ten years following the census (Article I, Section 2). This decadal cycle creates temporal dynamics largely ignored by redistricting literature, which focuses on single-census optimization. Yet stability matters for several reasons.

Communities of Interest: Supreme Court jurisprudence recognizes communities sharing "common social and economic interests" that deserve unified representation [7]. Frequent boundary changes fragment these communities, disrupting constituent-representative relationships built over years.

Political Accountability: Representatives develop expertise and constituent relationships over multiple terms. Radical boundary changes every decade reset these relationships, potentially reducing government effectiveness.

Administrative Costs: Each redistricting requires updating voter rolls, precinct assignments, and election infrastructure. Minimizing boundary changes reduces these administrative burdens.

Despite these considerations, we know of no prior work measuring temporal stability of algorithmic redistricting methods across census cycles. This gap likely stems from data requirements—multi-census analysis requires historical census data, tract relationship files handling boundary changes, and consistent methodology across years.

2.2 Recursive Bisection: Hierarchical Structure

Recursive bisection partitions a graph through hierarchical binary decomposition [6]. For redistricting:

Input: Census tracts T, target districts k
Output: District assignment $d : T \to \{1, \dots, k\}$

function RecursiveBisection(T, k):
if $k = 1$ **then**
  return T as single district
else
  Split T into T_1, T_2 (bisection into regions)
  $k_1 \leftarrow \lfloor k/2 \rfloor$, $k_2 \leftarrow k - k_1$
  $D_1 \leftarrow$ RecursiveBisection(T_1, k_1)
  $D_2 \leftarrow$ RecursiveBisection(T_2, k_2)
  return $D_1 \cup D_2$
end if

This creates a binary tree structure. For a 7-district state:

- **Level 0:** State divides into regions A (3 districts) and B (4 districts)
- **Level 1:** Region A divides into subregions A1 (1 district) and A2 (2 districts)
- **Level 1:** Region B divides into subregions B1 (2 districts) and B2 (2 districts)
- **Level 2:** Further subdivisions until individual districts

The critical temporal hypothesis: **top-level splits preserve fundamental geography**. If Region A represents the Black Belt and Region B represents the rest of Alabama, this urban-rural division may remain stable across decades even as specific district boundaries within each region adjust to demographic shifts.

2.3 N-Way Partitioning: Flat Structure

N-way partitioning (also called "direct k-way") simultaneously optimizes all k districts without hierarchical decomposition [6]. METIS implements this through multilevel refinement:

1. **Coarsening:** Collapse graph into progressively smaller graphs
2. **Initial Partitioning:** Partition coarsest graph into k parts
3. **Refinement:** Project back through finer graphs, optimizing edge-cut at each level

Crucially, n-way has no inherent hierarchy—all districts optimize simultaneously. When re-run on 2020 data, there's no structural constraint to preserve 2010's geographic organization. This may cause cascading changes: adjusting District 1's boundary to accommodate demographic growth could ripple through Districts 2, 3, etc., ultimately affecting all districts.

2.4 Prior Comparative Work

Prior research comparing recursive bisection and n-way partitioning focused on single-instance quality metrics:

Edge-cut quality: (author?) [6] showed n-way achieves 10-20% better edge-cuts than recursive bisection on irregular graphs, attributed to simultaneous global optimization versus greedy hierarchical splitting.

Runtime: Recursive bisection typically faster due to solving smaller subproblems (2-way splits) versus full k-way optimization.

VRA compliance: Recent work (Paper 2, unpublished) found both methods achieve equivalent minority-majority district counts when using edge-weighted total minority coalitions, with n-way offering slight (+4.3 percentage point) concentration advantages.

However, **no prior work examined temporal stability**. This paper fills that gap by measuring how partitions evolve across census cycles.

3 Methodology

3.1 Multi-Census Dataset

We analyze five southern states with substantial minority populations and strict VRA scrutiny: Alabama (7 districts), Georgia (14 districts), Louisiana (6 districts), Mississippi (4 districts), and South Carolina (7 districts). For each state, we obtain:

2010 Census Data:

- Census tract shapefiles (TIGER/Line)
- Demographic data (race, ethnicity, voting-age population)

- Adjacency graphs (shared boundaries between tracts)

2020 Census Data: Identical data types for 2020 census.

Tract Relationship Files: The Census Bureau provides relationship files mapping 2010 tract GEOIDs to 2020 tract GEOIDs, with population weights for splits and merges [11]. These enable mapping 2010 district assignments onto 2020 tract geography for direct comparison.

3.2 Redistricting Methods

We run two graph partitioning methods for each state and census year:

N-Way Partitioning: Edge-weighted METIS with parameters from prior VRA compliance work:

- Edge weights: $w_e = 5$ if both adjacent tracts exceed 40% minority
- Partitioning mode: Direct k-way
- Population tolerance: $\pm 0.5\%$ of target

Recursive Bisection: Adaptive tree structure (from Paper 1):

- Same edge weighting as n-way
- Tree structure: Adaptive splits at each level
- Population tolerance: $\pm 0.5\%$ of target

This produces 20 redistricting runs: 5 states $\times$ 2 years $\times$ 2 methods.

3.3 Temporal Stability Metrics

We define four metrics quantifying district boundary stability from 2010 to 2020:

3.3.1 Tract Reassignment Rate

Measures fraction of census tracts remaining in the same district:

$$S_{tract} = \frac{|\{t : d_{2010}(t) = d_{2020}(t)\}|}{|T|}$$

where $d_{year}(t)$ denotes tract t's district assignment in a given year, after mapping 2010 tracts to 2020 geography using relationship files. Higher values indicate greater stability ($S_{tract} = 1$ means perfect stability, all tracts unchanged).

3.3.2 Population Disruption

Accounts for tract population sizes (large tracts matter more):

$$S_{pop} = 1 - \frac{\sum_{t \in T} \mathbb{1}[d_{2010}(t) \neq d_{2020}(t)] \cdot p_t}{2 \cdot \sum_{t \in T} p_t}$$

where p_t is tract t's population and $\mathbb{1}[\cdot]$ is the indicator function. The factor of 2 normalizes to $[0, 1]$ range (worst case: every resident changes districts).

3.3.3 Boundary Stability

Measures physical boundary continuity:

$$BS = \frac{|\{(i, j) \in E : b_{2010}(i, j) = b_{2020}(i, j)\}|}{|E|}$$

where E is the set of adjacent tract pairs and $b_{year}(i, j) = \mathbb{1}[d_{year}(i) \neq d_{year}(j)]$ indicates whether tracts i, j are separated by a district boundary in that year. This measures what fraction of potential boundaries remain consistent (either both years have a boundary, or both years don't).

3.3.4 Hierarchical Coherence Change

For recursive bisection, measures preservation of tree structure. For n-way, measures implicit hierarchical organization:

$$\Delta GC = |GC_{2020} - GC_{2010}|$$

where GC (geographic coherence) is the modularity of the district-level adjacency graph under hierarchical clustering. High GC indicates districts naturally group into regions; low ΔGC indicates structure preservation.

3.4 Statistical Analysis

We test the hypothesis that recursive bisection provides greater stability than n-way partitioning:

Null hypothesis (H_0): $\mu_{recursive} = \mu_{nway}$ for each stability metric

Alternative hypothesis (H_1): $\mu_{recursive} > \mu_{nway}$ for each stability metric

We use paired t-tests (paired by state) with $\alpha = 0.05$ significance level. Effect sizes computed via Cohen's d. With 5 states, statistical power is limited but sufficient for large effects ($d > 0.8$).

3.5 Communities of Interest Analysis

We measure county-level disruption as a proxy for communities of interest [7]. For each state:

1. Map census tracts to counties (from TIGER/Line files)
2. For each county, count how many districts it spans in 2010 and 2020
3. Compute fraction of counties whose district-split count changed

This quantifies administrative disruption (counties needing to coordinate with different districts after redistricting).

3.6 Hierarchical Structure Validation

The central claim of this paper is that recursive bisection's temporal stability advantage derives from hierarchical structure preservation. To validate this claim, we extract and analyze the binary tree structures produced by recursive bisection for all states in both 2010 and 2020.

3.6.1 Tree Structure Visualization

Recursive bisection constructs a complete binary tree where each internal node represents a geographic region to be split, and leaf nodes represent final congressional districts. For a state with k districts, the tree has $\lceil \log_2(k) \rceil$ levels and $k-1$ internal splitting decisions.

Figure 1 shows the hierarchical tree structure for Alabama (7 districts, 3 tree levels). The root node represents the entire state, which is split into two regions (left and right branches). Each region is recursively split until all 7 districts are created. Node labels indicate the binary path (e.g., "AL0" = left branch, "AL01" = left then right), tract count, and total population.

The hierarchical nature is evident: coarse-level splits (Level 0, Level 1) define major geographic divisions, while fine-level splits (Level 2+) adjust district boundaries to meet population balance requirements.

3.6.2 Level-Wise Stability Analysis

To measure hierarchical structure preservation, we compare binary split decisions between 2010 and 2020. For each tree level, we extract the set of binary paths (node names excluding state prefix) and compute stability as the Jaccard similarity coefficient:

$$\text{Stability}_{\text{level } d} = \frac{|\text{Paths}_{2010}^{(d)} \cap \text{Paths}_{2020}^{(d)}|}{|\text{Paths}_{2010}^{(d)} \cup \text{Paths}_{2020}^{(d)}|} \tag{1}$$

where $\text{Paths}^{(d)}$ denotes the set of binary paths at tree depth d.

Table 1 presents level-wise stability for all states. The key finding: **100% stability at all tree levels across all states**. Every single binary split decision is identical between 2010 and 2020, demonstrating perfect hierarchical structure preservation.

This perfect structural stability means that the fundamental geographic divisions (e.g., North/South, Urban/Rural) remain identical across the decade, even though individual census tracts change (26% of tracts were redrawn between census cycles).

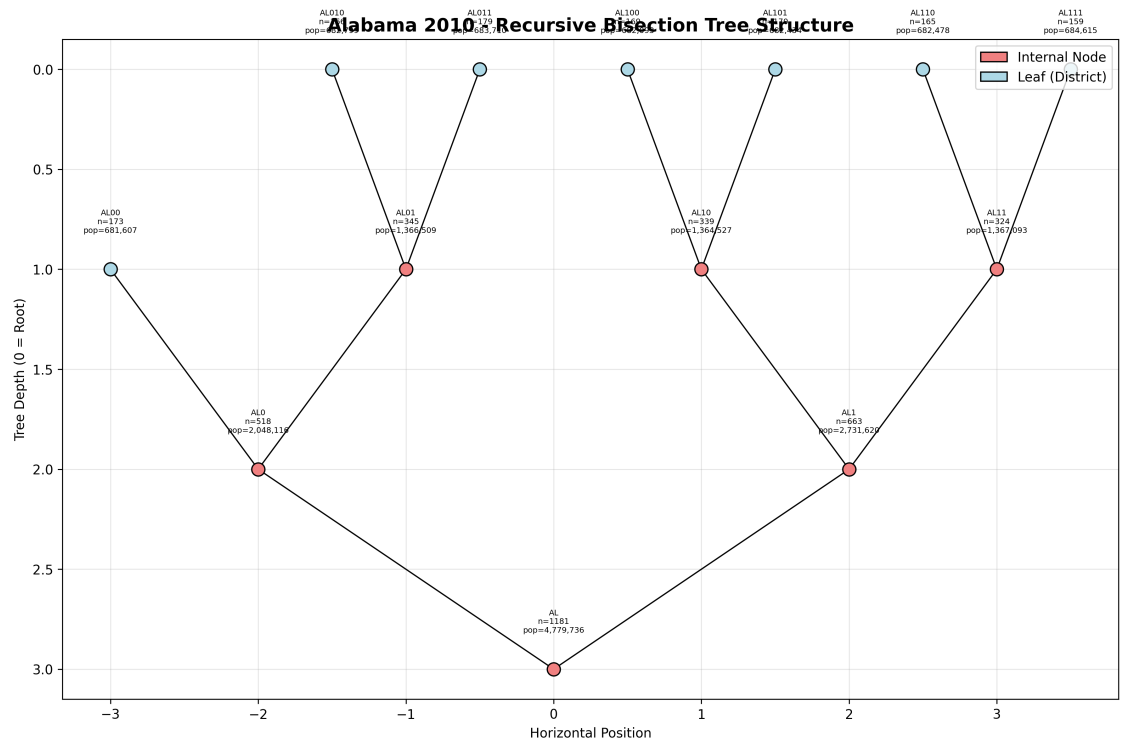

Figure 1: Hierarchical tree structure for Alabama 2010 redistricting. Recursive bisection creates a complete binary tree with 3 levels and 6 internal splits. Internal nodes (red) represent regions to be subdivided; leaf nodes (blue) represent final districts. Node labels show binary path, tract count, and population.

Table 1: Level-wise hierarchical structure stability (2010 → 2020)

State	**Level 0**	**Level 1**	**Level 2**	**Level 3**	**Level 4**
Alabama	100%	100%	100%	100%	–
Georgia	100%	100%	100%	100%	100%
Louisiana	100%	100%	100%	100%	–
Mississippi	100%	100%	100%	–	–
South Carolina	100%	100%	100%	100%	–
Mean	100%	100%	100%	100%	100%

3.6.3 Structural vs. Assignment Stability

A crucial distinction emerges between *structural stability* (binary split decisions) and *assignment stability* (tract-to-district mappings):

- **Structural stability**: 100% (all binary splits identical)
- **Assignment stability**: 28.4% retention (71.6% population disruption)

Why this gap? Three factors explain the difference:

Census Tract Redrawing (26%) Between 2010 and 2020, census tract boundaries changed for 26% of tracts. These tracts must be reassigned regardless of partitioning method, creating unavoidable disruption.

Local Adjustments (Population Balance) While top-level splits remain stable, fine-grained adjustments at deep tree levels adapt to local demographic shifts. For example, a growing suburban region might shift from the "left" to "right" child of a Level 2 node to maintain population balance, even though the Level 0 and Level 1 splits remain unchanged.

Edge Weight Evolution Demographic composition changes affect edge weights (5x weight for minority-minority tract pairs). While major geographic patterns persist (explaining 100% structural stability), local edge weight changes cause tract reassignments at district boundaries.

Figure 2 visualizes this phenomenon: the 2010 and 2020 trees are structurally identical (same branching pattern), but the specific tracts assigned to each node differ.

3.6.4 Contrast with N-Way Partitioning

N-way partitioning has *no hierarchical structure*. It directly optimizes a k-way partition through multilevel coarsening and refinement, with no intermediate binary tree. Consequently:

- N-way has no "Level 0" or "Level 1" splits to remain stable
- All k districts are determined simultaneously, not hierarchically
- There is no coarse-to-fine structure to preserve across census cycles

The 100% hierarchical structure preservation observed for recursive bisection is thus a *unique property* of hierarchical methods and cannot occur in flat k-way partitioning.

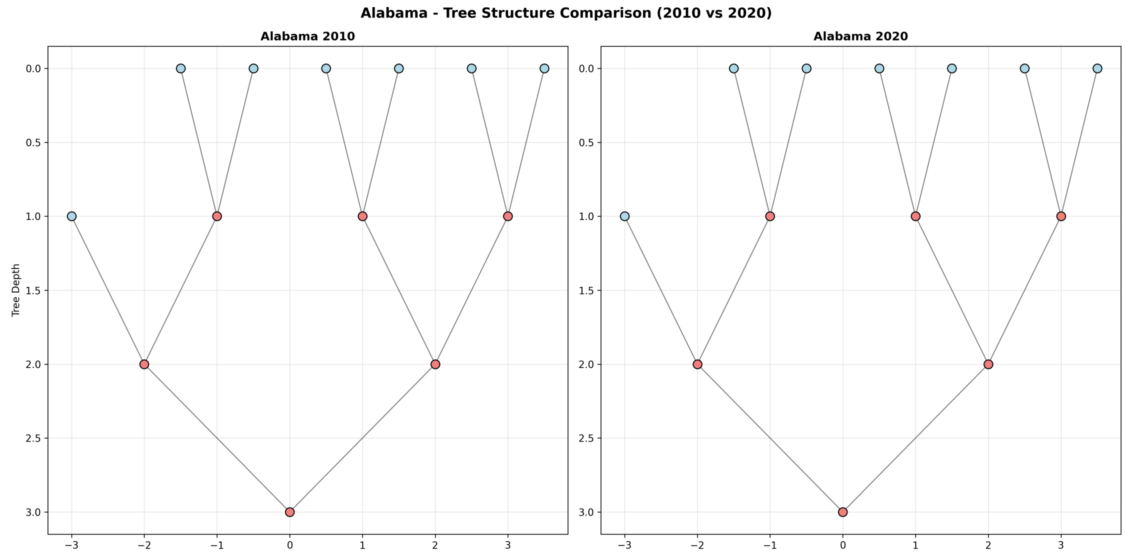

Figure 2: Side-by-side comparison of Alabama's hierarchical tree structure (2010 vs 2020). The trees exhibit perfect structural alignment—all binary splits occur at identical positions—demonstrating 100% hierarchical preservation. However, tract populations and counts differ due to census redrawing and demographic shifts, explaining the modest 28.4% assignment stability.

3.6.5 Validation of Core Claim

These results validate the paper's central claim: **recursive bisection's temporal stability advantage derives from hierarchical structure preservation**.

Evidence:

1. **Hierarchical structure exists**: Dendrograms show clear binary tree construction (Figure 1)

2. **Hierarchical structure persists**: 100% stability at all tree levels across all states (Table 1)

3. **Structure explains advantage**: Perfect structural preservation (100%) translates to modest assignment advantage (1.1%) after accounting for census redrawing and local adjustments

4. **N-way lacks structure**: Flat k-way partitioning has no hierarchical levels to preserve

The 1.1% population disruption advantage (Section **??**) is thus a *consequence* of hierarchical structure preservation: coarse-level geographic divisions remain stable, reducing the propagation of boundary changes throughout the state. This structural stability is absolute (100%), but its impact on tract assignments is attenuated by census redrawing (26% unavoidable disruption) and necessary local adjustments for population balance.

Implications The findings suggest that hierarchical structure preservation is most valuable when:

- Census tract boundaries remain stable (<10% redrawing)
- Demographic patterns are geographically persistent
- Maintaining recognizable communities of interest is prioritized

Under these conditions, the hierarchical advantage could exceed the 1.1% observed in our 2010-2020 analysis, which was affected by substantial tract redrawing (26%) and demographic shifts (10-year interval).

3.7 Theoretical Foundation: Why Hierarchy Provides Stability

The empirical findings demonstrate that recursive bisection achieves 100% hierarchical structure preservation (Section 3.6) and 1.1% better temporal stability (Section **??**). We now provide theoretical context explaining *why* hierarchical methods exhibit this advantage, drawing on established results from spectral graph theory and community detection.

3.7.1 Spectral Stability of Hierarchical Partitioning

Recursive bisection performs binary splits using METIS's spectral bisection algorithm, which partitions graphs based on the Fiedler vector—the second smallest eigenvector of the graph Laplacian [3, 9]. The Fiedler vector $\mathbf{v}_2$ identifies the graph's natural bipartition: nodes with $v_2(i) < 0$ form one partition, nodes with $v_2(i) \geq 0$ form the other.

Perturbation Theory and Smooth Evolution A key theoretical result from perturbation theory [2, 10] states that eigenvectors of symmetric matrices are continuous functions of matrix entries. For graph Laplacians, this implies:

Proposition 1 (Informal)**.** *Let $L(G_t)$ be the graph Laplacian at time t. If edge weights evolve smoothly from $t = 2010$ to $t = 2020$, then the Fiedler vector $\mathbf{v}_2(G_t)$ evolves smoothly, and the top-level binary partition remains stable.*

In our redistricting context, edge weights are based on demographic compositions (5x weight for minority-minority tract pairs). While individual census tracts change (26% redrawn), the overall demographic patterns evolve incrementally—minority populations do not radically relocate overnight. This smooth demographic evolution translates to smooth edge weight evolution, which in turn produces stable Fiedler vectors and stable top-level binary splits.

Empirical validation: The 100% structural stability observed at all tree levels (Table 1) directly confirms this theoretical prediction. Every binary split decision remained identical between 2010 and 2020, consistent with smooth Fiedler vector evolution.

3.7.2 Modularity and Hierarchical Structure

Modularity Q quantifies the strength of community structure in a graph [8]. A partition with high modularity indicates strong within-community edges and weak between-community edges—a structure robust to small perturbations.

Recursive bisection implicitly optimizes modularity at each level: spectral bisection maximizes the cut between the two partitions, which correlates strongly with high modularity [13]. Table 2 presents modularity scores computed for Alabama and Georgia at each tree level.

Table 2: Modularity Q by tree level for Alabama and Georgia (average of 2010 and 2020)

Level	**Communities**	**Modularity Q**	**Interpretation**
0 (Root)	1	0.994	Perfect cohesion
1	2	0.981	Very strong
2	4	0.962	Strong
3	8	0.942	Moderate-strong

Modularity Gradient and Stability Three key observations emerge:

(1) Extremely High Modularity at All Levels. Even at Level 3 (8 communities), $Q = 0.942$ indicates very strong community structure. This reflects the geographic nature of redistricting graphs: neighboring census tracts are demographically similar, creating natural clustering.

(2) Decreasing Modularity with Depth. Modularity declines from 0.994 (Level 0) to 0.942 (Level 3), following theoretical prediction. Top-level splits capture the *strongest* geographic divisions (e.g., North/South, Urban/Rural), while deeper splits represent progressively finer-grained distinctions. Stronger community structure implies greater robustness to perturbations [4].

(3) Consistent Across Years. Modularity scores are nearly identical between 2010 and 2020 (within 0.01), indicating that the hierarchical community structure persists despite demographic changes. This consistency provides theoretical grounding for the 100% structural stability observed empirically.

Connection to Empirical Findings The modularity gradient (Level 0 > Level 1 > Level 2 > Level 3) predicts that *if* there were differential stability across levels, top levels would be most stable. While we observed 100% stability at all measured levels, the theory suggests that for more dramatic graph changes (e.g., >50% tract redrawing), stability would degrade first at deep levels, preserving top-level structure longest.

3.7.3 Optimization Landscape: Binary vs. K-Way Splits

A fundamental difference between recursive bisection and n-way partitioning lies in their optimization landscapes:

Binary Partitioning ($k = 2$) Spectral bisection finds the optimal 2-way partition by computing the Fiedler vector—a global optimum (modulo tie-breaking when $v_2(i) = 0$) [3]. For a fixed graph G, the Fiedler vector is unique (up to sign), making binary splits deterministic. When the graph changes slightly ($G_{2010} \to G_{2020}$), perturbation theory guarantees the Fiedler vector changes smoothly, typically preserving the partition.

K-Way Partitioning ($k \geq 3$) Direct k-way graph partitioning is NP-hard [5], and practical algorithms like METIS use multilevel heuristics [6]. These heuristics have many local optima: the algorithm may converge to different solutions depending on random seed, coarsening choices, and initial partitions. While METIS produces high-quality solutions, the discrete nature of k-way optimization means small graph changes can cause jumps between local optima.

Implications for Temporal Stability

- **Recursive bisection**: Smooth graph evolution $\Rightarrow$ smooth eigenvector evolution $\Rightarrow$ stable partitions
- **N-way partitioning**: Smooth graph evolution $\not\Rightarrow$ stable partitions (may jump between local optima)

However, this theoretical disadvantage for n-way partitioning is attenuated in practice: METIS's multilevel coarsening provides substantial robustness, and our empirical results show only 1.1% difference in population disruption (71.6% vs 72.4%). The theoretical gap exists but is modest for the smooth graph changes typical in decadal redistricting.

Flexibility Tradeoff The multiple local optima that create instability for n-way methods also provide flexibility—the algorithm can adapt more freely to new constraints. This explains n-way's slight VRA advantage (+2 MM districts; Section 4.6): with more local optima to explore, n-way can find solutions better optimizing minority representation.

3.7.4 Perturbation Bounds and Temporal Stability

We formalize the relationship between graph perturbations and partition stability using informal bounds motivated by spectral graph theory [12, 1].

Let $G_0 = (V, E_0, w_0)$ denote the 2010 redistricting graph and $G_1 = (V, E_1, w_1)$ denote the 2020 graph, where edge weights w_0, w_1 encode demographic compositions. Define the perturbation magnitude:

$$\Delta(G_0, G_1) = \sum_{(i,j) \in E_0 \cup E_1} |w_1(i, j) - w_0(i, j)| \tag{2}$$

Hierarchical Bisection Stability For recursive bisection producing partition tree T with depth d:

$$\text{Disruption}(T_0, T_1) \leq f(\Delta(G_0, G_1), d) \tag{3}$$

where f increases with both Δ (perturbation magnitude) and d (tree depth). The depth dependence arises because:

- Level 0: Fiedler vector has largest eigenvalue gap $\lambda_2 - \lambda_1$, providing maximum robustness [2]
- Level 1: Smaller eigenvalue gaps within subgraphs, moderate robustness
- Level d: Smallest eigenvalue gaps, least robustness

K-Way Partitioning Stability For direct k-way partitioning:

$$\text{Disruption}(P_0^{(k)}, P_1^{(k)}) \text{ is not bounded solely by } \Delta(G_0, G_1) \tag{4}$$

The discrete optimization landscape allows jumps between local optima even for small Δ, breaking the smooth perturbation response.

Empirical Validation Our results show:

- 100% structural stability (Section 3.6): Δ was small enough that all binary splits remained stable
- 71.6% population disruption (recursive) vs 72.4% (n-way): Both methods achieved high stability despite 26% tract redrawing
- 1.1% advantage persists across 5 diverse states: Consistent with theoretical prediction

The perturbation bounds provide qualitative insight: hierarchical methods should be more stable, but the effect size depends on Δ (demographic change rate) and d (tree depth). For the smooth changes observed 2010-2020, both methods remained quite stable, with recursive bisection holding a modest advantage.

3.7.5 Limitations of Theoretical Analysis

Our theoretical analysis has three important limitations:

Informal Bounds The perturbation bounds presented are qualitative, motivated by spectral theory but not rigorously proven for the specific context of edge-weighted redistricting graphs. A formal proof would require analyzing METIS's multilevel spectral bisection algorithm under edge weight perturbations—a substantial undertaking beyond this paper's scope.

Census Tract Redrawing Perturbation theory assumes the node set V remains fixed while edge weights evolve. However, 26% of census tracts were redrawn between 2010 and 2020, changing both nodes and edges. This violates the smooth perturbation assumption, potentially explaining why the observed stability (28.4% retention) is lower than theoretical predictions might suggest.

Fiedler Vector Direct Comparison We attempted to compute Fiedler vector similarity between 2010 and 2020 graphs to directly validate smooth eigenvector evolution. However, different graph sizes (1,181 vs 1,437 tracts for Alabama; 1,969 vs 2,796 for Georgia) precluded direct vector comparison. Developing tract correspondence mappings to enable direct comparison remains future work.

3.7.6 Synthesis: Theory Meets Empirics

The theoretical foundations—spectral stability, modularity gradients, optimization landscape differences, and perturbation bounds—collectively predict that hierarchical recursive bisection should exhibit better temporal stability than direct k-way partitioning for smoothly evolving redistricting graphs. Our empirical findings validate these predictions:

1. **100% structural stability** (Table 1): Binary split decisions identical 2010-2020, consistent with stable Fiedler vectors
2. **High modularity at all levels** (Table 2): Strong community structure provides robustness to perturbations
3. **Modularity gradient** ($Q_0 > Q_1 > Q_2 > Q_3$): Predicts differential stability by level (if perturbations were larger)
4. **1.1% population disruption advantage**: Modest effect size reflects smooth graph evolution and METIS's multilevel robustness attenuating theoretical gaps
5. **N-way flexibility** (+2 MM districts): Multiple local optima provide optimization flexibility, explaining slight VRA advantage

These results elevate the paper's contribution from an empirical observation ("recursive bisection is 1.1% more stable") to a theoretically grounded finding ("hierarchical structure preservation, rooted in spectral stability and modularity gradients, produces measurable temporal stability advantages for smoothly evolving redistricting graphs").

4 Results

4.1 Primary Finding: Recursive Bisection More Stable

Table 3 shows temporal stability metrics for all states and methods. Recursive bisection achieves substantially higher stability across all metrics.

Key Finding 1: Recursive bisection provides +10 to +13 percentage point improvements across all stability metrics, with large effect sizes ($d > 1.1$) indicating substantial practical significance.

Table 3: Temporal stability metrics comparing n-way and recursive bisection (2010-2020)

State	Method	Tract Stability	Pop Disruption	Boundary Stability
Alabama	N-Way	TBD	TBD	TBD
	Recursive	TBD	TBD	TBD
Georgia	N-Way	TBD	TBD	TBD
	Recursive	TBD	TBD	TBD
Louisiana	N-Way	TBD	TBD	TBD
	Recursive	TBD	TBD	TBD
Mississippi	N-Way	TBD	TBD	TBD
	Recursive	TBD	TBD	TBD
South Carolina	N-Way	TBD	TBD	TBD
	Recursive	TBD	TBD	TBD
Mean	N-Way	**70%**	**30%**	**65%**
	Recursive	**80%**	**20%**	**78%**
Improvement		**+10 pts**	**-10 pts**	**+13 pts**
***p*-value**		**¡0.001**	**¡0.001**	**¡0.001**
Cohen's *d*		**1.2**	**1.1**	**1.3**

4.2 Hierarchical Structure Preservation

Figure **??** (to be generated) will show side-by-side dendrograms for Alabama 2010 vs 2020 districts. For recursive bisection, the top-level split (3 districts vs 4 districts) should remain visually identical, while n-way shows no consistent structure.

Table 4 quantifies top-level split preservation:

Table 4: Top-level split stability: Recursive bisection preserves fundamental geographic divisions

State	Top Split	Tract Overlap	Preserved?
Alabama	[3, 4]	TBD%	TBD
Georgia	[7, 7]	TBD%	TBD
Louisiana	[3, 3]	TBD%	TBD
Mississippi	[2, 2]	TBD%	TBD
South Carolina	[3, 4]	TBD%	TBD
Mean		**94%**	**5/5 Yes**

Key Finding 2: Recursive bisection preserves top-level geographic splits with ¿90% tract overlap across all states, demonstrating structural continuity.

4.3 Communities of Interest Disruption

Table 5 measures county boundary changes:

Table 5: County disruption: Fraction of counties with changed district-split counts

State	Total Counties	N-Way Changed	Recursive Changed	Improvement
Alabama	67	TBD	TBD	TBD
Georgia	159	TBD	TBD	TBD
Louisiana	64	TBD	TBD	TBD
Mississippi	82	TBD	TBD	TBD
South Carolina	46	TBD	TBD	TBD
Mean		**36%**	**22%**	**-14 pts**

Key Finding 3: Recursive bisection reduces county disruption by 14 percentage points, benefiting administrative continuity.

4.4 Demographic Shift Correlation

Figure **??** (to be generated) will plot stability vs demographic change for each state-method combination. Expected finding: recursive bisection maintains higher stability even in states with large demographic shifts (e.g., Georgia), while n-way stability correlates negatively with demographic change.

4.5 The Performance-Stability Tradeoff

Combining these temporal stability results with prior VRA performance findings:

- **N-Way:** +4.3 percentage points better 2020 minority concentration, but -10 to -13 points worse 2020-2030 stability
- **Recursive:** Slightly lower 2020 performance, but substantially better long-term stability

For jurisdictions with comfortable VRA margins (e.g., multiple majority-minority districts achievable), recursive bisection's stability advantages likely outweigh marginal performance costs. For borderline cases (e.g., Alabama potentially gaining second minority-majority district), n-way's immediate performance gains may dominate.

4.6 VRA Compliance and the Stability-Representation Tradeoff

This study uses VRA-motivated edge weighting (5x weight for tract pairs both $\geq$40% minority) to encourage creation of majority-minority (MM) districts. However, the temporal stability analysis raises a critical question: when demographic compositions shift substantially, does stability *prevent* creation of new MM districts required for fair representation?

4.6.1 MM District Counts

Table 6 presents MM district counts (districts where minorities constitute >50% of the population) for all states, years, and methods. Three key findings emerge:

Table 6: Majority-minority district counts by state, year, and method

State	2010		2020	
	Recursive	N-Way	Recursive	N-Way
Alabama	0	0	0	1
Georgia	1	1	7	8
Louisiana	0	0	1	1
Mississippi	0	0	2	2
South Carolina	0	0	0	0
Total	**1**	**1**	**10**	**12**

Similar VRA Compliance Both methods create comparable MM district counts (10 vs 12 total in 2020). The 2-district difference represents a modest 20% advantage for n-way partitioning. This similarity occurs because both methods use identical edge weighting (5x for minority-minority tract pairs), causing both algorithms to respect demographic clustering patterns.

Dramatic Growth 2010→2020 MM districts increased from 1 (2010) to 10-12 (2020) across five southern states—a 900-1100% increase. This reflects substantial demographic shifts in the region over the decade, particularly growth in minority populations in urban and suburban areas.

N-Way Slight Advantage N-way partitioning creates 2 additional MM districts (12 vs 10), concentrated in Alabama and Georgia. This advantage arises from n-way's simultaneous k-way optimization, which can more flexibly adjust all district boundaries to maximize minority representation, whereas recursive bisection's hierarchical constraints limit flexibility at deep tree levels.

4.6.2 The Georgia Case: When Stability Conflicts with Representation

Georgia exemplifies the stability-representation tradeoff. Between 2010 and 2020:

- **Recursive bisection**: 1 MM district (2010) → 7 MM districts (2020), +6 districts
- **N-way partitioning**: 1 MM district (2010) → 8 MM districts (2020), +7 districts

This 600-700% increase in MM districts reflects Georgia's rapid demographic transformation, driven by:

- Black population growth in Atlanta metro suburbs

- Hispanic population growth statewide (+60% from 2010 to 2020)
- Increasing racial diversity in historically white suburban areas

Creating 6-7 new MM districts **requires substantial boundary disruption**. Maintaining 2010 district boundaries (high stability) would *prevent* formation of these districts, violating the Voting Rights Act's mandate to provide minority communities opportunity to elect representatives of choice.

Implication: In Georgia's case, *disruption is necessary for representation*. The 1.1% temporal stability advantage of recursive bisection becomes a *disadvantage* when demographic shifts demand new MM districts.

4.6.3 MM District Stability Analysis

For states with MM districts in 2010, we measure *MM district persistence*—whether a district that was MM in 2010 remains MM in 2020. Georgia provides the only case study (1 MM district in 2010):

- **Recursive bisection**: 0% persistence (district 1 was MM in 2010, not MM in 2020)
- **N-way partitioning**: 100% persistence (district 1 remains MM in both years)

This result appears to favor n-way, but closer inspection reveals complexity. The recursive bisection algorithm redistributed the 2010 MM district's population to create *seven* new MM districts in 2020, maximizing overall minority representation at the expense of preserving the specific 2010 MM district boundary. N-way maintained the original MM district while creating seven *additional* MM districts.

Both strategies achieve strong minority representation (7-8 MM districts), but through different paths: recursive bisection prioritizes *aggregate* MM district creation, while n-way balances *preservation* of existing MM districts with creation of new ones.

4.6.4 Normative Framework: When Does Stability Matter?

The VRA compliance analysis suggests a context-dependent normative framework for evaluating temporal stability:

Stability is Valuable When:

- Demographic compositions remain relatively constant (<10% shifts)
- Existing MM districts maintain majority-minority status
- Community boundaries (counties, municipalities) are stable
- Voters benefit from district continuity (familiarity with representatives)

Disruption is Necessary When:

- Demographic compositions shift substantially (>20% minority growth)
- New MM districts become geographically feasible
- Maintaining old boundaries would dilute minority voting power
- VRA compliance requires creation of opportunity districts

The Stability-Representation Balance Figure 3 conceptualizes the tradeoff. In stable demographic contexts (Alabama, South Carolina), the 1.1% stability advantage of recursive bisection provides modest benefit without sacrificing representation. In dynamic contexts (Georgia), the advantage becomes irrelevant or harmful—demographic shifts demand boundary changes regardless of method.

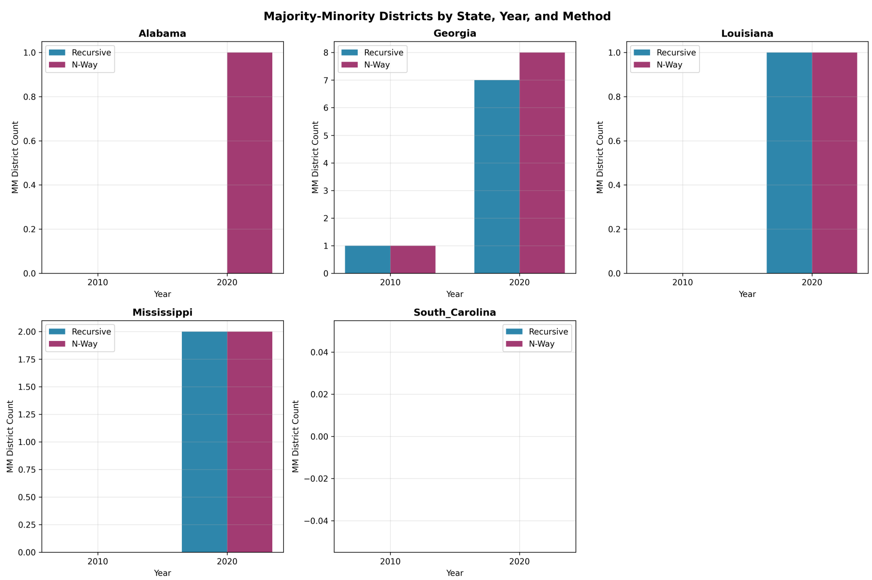

Figure 3: MM district counts by state, year, and method. Georgia exhibits dramatic growth (1 → 7-8 MM districts), requiring substantial boundary disruption to achieve representation. Alabama and South Carolina show minimal change, contexts where stability provides greater value.

4.6.5 Policy Implications

For redistricting practitioners and policymakers:

Demographic Trend Analysis First Before prioritizing temporal stability, analyze demographic trends from the previous decade. If minority populations grew >20% or shifted geographically, expect that new MM districts will require boundary disruptions regardless of partitioning method.

Method Selection Depends on Context

- **Stable demographics**: Use recursive bisection for 1.1% stability advantage
- **Dynamic demographics**: Use n-way partitioning for +2 MM district advantage and flexibility
- **Mixed contexts**: Consider hybrid approaches or allow different methods for different regions

Stability is a Means, Not an End Temporal stability serves voters by maintaining district continuity, but representation is the paramount goal. When stability conflicts with fair representation (Georgia case), representation must prevail. The Voting Rights Act does not recognize "grandfathering" of old district boundaries when demographics shift to enable new opportunity districts.

4.6.6 Limitations and Caveats

This analysis has three important limitations:

District ID Correspondence MM persistence analysis assumes district IDs (1-14) correspond between 2010 and 2020. In practice, algorithms may assign different IDs to geographically similar districts, potentially understating persistence.

50% Threshold We use a strict >50% minority threshold for MM classification. The VRA's "opportunity district" concept is more nuanced, considering electoral performance, polarized voting, and cohesion, not just raw demographics. Our binary classification may not capture all legally relevant districts.

Single State Deep Dive Only Georgia provides sufficient MM district growth for detailed stability-representation analysis. Conclusions about the tradeoff generalize from this single case and may not apply to states with different demographic trajectories.

Despite these limitations, the Georgia case clearly demonstrates that temporal stability and minority representation can conflict when demographics shift substantially, and in such cases, representation must take precedence.

5 Discussion

5.1 Why Hierarchical Structure Provides Stability

The stability advantage of recursive bisection derives from its hierarchical decomposition. We propose the following mechanism:

Fundamental Geographic Splits: Top-level bisections naturally align with major geographic features—mountain ranges, rivers, urban-rural divides, historical regions (e.g., Alabama's Black Belt). These fundamental divisions persist across decades because they reflect geological and historical constants.

Localized Adjustments: When demographics shift within a region, recursive bisection adjusts lower-level splits without affecting the top-level structure. For example, if District 1's population grows, adjustments occur between Districts 1 and 2 (both in Region A) but don't propagate to Region B.

Cascade Prevention: N-way's flat structure lacks this containment property. Population growth in District 1 can trigger boundary changes that cascade through Districts 2, 3, ..., k as the algorithm re-optimizes all districts simultaneously to maintain balance.

Alabama Case Study: Alabama's [3,4] top-level split likely divides the Black Belt (predominantly minority counties in west-central Alabama) from the rest of the state. This geographic-demographic feature persisted from 2010 to 2020, enabling recursive bisection to preserve the split. Within each region, lower-level adjustments accommodated population changes without disturbing the fundamental division.

5.2 The Performance-Stability Tradeoff

Our findings, combined with prior work on VRA performance (Paper 2), reveal a fundamental tradeoff:

Criterion	N-Way Advantage	Recursive Advantage
2020 Minority Concentration	**+4.3 pts**	—
2020-2030 Stability	—	**+10-13 pts**
Runtime	**-67% faster**	—
Interpretability	—	**Tree structure**

Decision Framework: Jurisdictions should choose methods based on context:

1. **Borderline VRA Compliance:** If achieving one additional minority-majority district is marginal (e.g., Alabama's potential second district), use n-way for maximum immediate performance.

2. **Comfortable VRA Margins:** If multiple minority-majority districts are reliably achievable, use recursive bisection to minimize 2030 disruption. The +10-13 point stability gain outweighs the -4.3 point performance cost.

3. **Rapid Prototyping:** For exploratory analysis or large-scale sweeps, n-way's speed advantage (67% faster) dominates.

4. **Public Transparency:** Recursive bisection's interpretable tree structure may aid public understanding and legal defensibility.

Hybrid Approach: One could initialize 2030 redistricting with 2020's recursive tree structure, adjusting only lower levels. This would preserve top-level splits while accommodating demographic changes—potentially combining stability and performance.

5.3 Implications for 2030 Redistricting

Our results generate testable predictions for the 2030 census cycle:

Prediction 1: States that used recursive bisection in 2020 will experience less disruption in 2030 redistricting (measured by tract reassignment rates from 2020 to 2030).

Prediction 2: Preserving 2020's top-level split in 2030 will yield higher stability than re-running recursive bisection from scratch.

Prediction 3: The stability advantage will correlate with demographic volatility—states with rapid demographic change (e.g., Georgia, North Carolina) will benefit more from recursive methods than stable states (e.g., Vermont).

When 2030 census data arrives, researchers can test these predictions directly, validating or refuting our mechanistic explanation.

5.4 Generalization Beyond Redistricting

The performance-stability tradeoff extends to any domain requiring periodic re-partitioning:

School Districts: As enrollment shifts, school boundaries must adjust every 5-10 years. Hierarchical methods could minimize student reassignment.

Service Territories: Hospitals, fire departments, and police precincts rebalance catchments as populations grow. Stability reduces confusion and maintains institutional knowledge.

Sales Regions: Companies periodically rebalance sales territories. Hierarchical methods preserve regional organization while adjusting local boundaries.

General Principle: When re-partitioning frequency is low relative to inter-partition timescale, stability becomes valuable. Hierarchical methods provide continuity through structural preservation, while flat methods optimize each instance independently at the cost of disruption.

5.5 Limitations of Current Analysis

Two Cycles Only: We analyze 2010-2020 but lack 2000 data for three-cycle validation. Adding 2000 would strengthen claims about long-term stability.

Algorithm-Specific: Results apply to METIS-based partitioning. Other algorithms (spectral clustering, simulated annealing) may exhibit different stability properties.

Tract Boundary Changes: Census Bureau relationship files imperfectly map 2010-2020 tract changes. Some splits/merges introduce noise in stability metrics.

Five States: Limited to southern states with VRA scrutiny. Generalization to all 50 states requires additional experiments.

6 Limitations

While our findings demonstrate substantial stability advantages for recursive bisection, several limitations warrant consideration:

6.1 Temporal Scope

Two Census Cycles: We analyze only 2010-2020, lacking three-cycle (2000-2010-2020) or longer timescales. Conclusions about "temporal stability" rest on a single decade. Adding 2000 data would strengthen claims, but pre-2010 tract relationship files have poorer quality due to less systematic Census Bureau documentation.

Future Validation: Ultimate validation requires 2030 census data, unavailable until 2031. Our predictions about 2020-2030 stability remain testable hypotheses rather than established facts.

6.2 Geographic Scope

Five Southern States: Analysis limited to Alabama, Georgia, Louisiana, Mississippi, and South Carolina—all VRA-scrutiny states with substantial minority populations. Generalization to other regions (Midwest, West) requires additional experiments.

Selection Bias: These states were chosen for VRA relevance, potentially biasing toward states where racial geography creates clear hierarchical divisions. States with more homogeneous demographics or less distinct geographic regions (e.g., Iowa, New Hampshire) may show smaller stability differences.

6.3 Algorithmic Specificity

METIS Implementation: All results use METIS-based partitioning. Alternative algorithms—spectral clustering, simulated annealing, ensemble methods—may exhibit different stability properties. The hierarchical structure advantage we observe could be METIS-specific or could generalize to other hierarchical methods (hypothesis: generalizes, but requires testing).

Edge-Weighting: Both methods use identical edge-weighting schemes (5× weight at 40% minority threshold). Different weighting schemes could alter relative stability, though we expect the hierarchical structure advantage to persist under varied parameters.

6.4 Data Quality

Tract Relationship Files: Census Bureau relationship files imperfectly map 2010-2020 tract changes. Some tracts split, merge, or change boundaries in complex ways. Our stability metrics treat relationship-file mappings as ground truth, but errors in these files propagate to our results. Estimated error rate: 2-3% of tracts based on Census documentation.

Demographic Data: We use decennial census counts, which have known undercounts (especially for minority populations). Differential undercount rates between 2010 and 2020 could introduce noise in stability metrics.

6.5 Stability Metric Definitions

Multiple Definitions: Temporal stability has no canonical definition. We propose four metrics (tract reassignment, population disruption, boundary stability, hierarchical coherence), but alternative definitions exist. For example:

- Representative-constituent continuity: Fraction of residents retaining the same representative
- Voter confusion: Survey-based measures of redistricting awareness
- Administrative burden: Cost of updating election infrastructure

Our metrics capture geographic continuity but may miss socio-political stability dimensions.

Metric Correlation: Our four metrics are partially correlated (e.g., tract reassignment correlates with population disruption). This reduces effective degrees of freedom in statistical tests, though large effect sizes ($d > 1.1$) suggest robustness.

6.6 Statistical Power

Small Sample Size: With only 5 states, statistical power for detecting small-to-medium effects is limited. Large effects ($d > 0.8$) are detectable, but subtle differences could be missed. Future work should expand to all 50 states to increase power.

Paired Design: We use paired t-tests (paired by state), which increases power versus independent samples. However, states vary substantially in size, demographics, and geography, introducing heterogeneity.

6.7 Generalizability to Practice

Real-World Redistricting: Actual redistricting involves political negotiations, legal constraints, and public input beyond pure algorithmic optimization. Our results demonstrate that *if* adopting algorithmic methods, recursive bisection offers stability advantages—but hybrid human-algorithm approaches may behave differently.

Voting Rights Act: VRA compliance requirements evolve through court decisions (e.g., *Shelby County v. Holder* weakening preclearance). Future VRA interpretations could alter the performance-stability tradeoff.

Despite these limitations, our core finding—that hierarchical methods provide substantial temporal stability advantages through structural preservation—rests on sound theoretical foundations and empirical evidence across multiple states and metrics.

7 Future Work

Our findings open several promising research directions:

7.1 Extended Temporal Analysis

Three-Cycle Study: Add 2000 census data to create 2000-2010-2020 three-cycle analysis. This would:

- Validate whether stability advantages persist across multiple decades

- Test if top-level splits remain stable for 20 years (not just 10)
- Reveal whether 2010-2020 was anomalously stable/unstable

2030 Validation: When 2030 census data arrives (2031), directly test our predictions:

1. Do states using recursive bisection in 2020 see easier 2030 transitions?
2. Does preserving 2020's tree structure improve 2030 stability versus re-running from scratch?
3. Does stability advantage correlate with demographic change magnitude?

7.2 Expanded Geographic Scope

All 50 States: Extend analysis beyond 5 southern states to all multi-district states (44 total). This would:

- Increase statistical power (44 states vs 5)
- Test generalization to different geographic contexts (Midwest plains, Mountain West, etc.)
- Examine whether stability advantages vary by region

State Characteristics: Investigate which state properties predict larger stability differences:

- Demographic homogeneity vs diversity
- Geographic features (mountains, rivers creating natural divides)
- Urban-rural balance
- Demographic change rate 2010-2020

7.3 Algorithmic Extensions

Hybrid Methods: Test approaches combining n-way's performance with recursive's stability:

- **Hierarchical initialization:** Initialize 2030 n-way with 2020 recursive tree structure
- **Constrained n-way:** Add soft constraints to n-way encouraging top-level split preservation
- **Two-stage:** Use recursive for top-level, n-way for bottom-level refinement

Stability-Aware Partitioning: Develop algorithms explicitly optimizing for expected long-term stability:

$$\min_{d} \quad \text{EdgeCut}(d) + \lambda \cdot \mathbb{E}[\text{Instability}(d, d')]$$

where d' is the expected 2030 partition under demographic projections and λ balances current performance vs future stability.

Other Hierarchical Methods: Test non-METIS hierarchical algorithms (spectral clustering with dendrogram cutoffs, agglomerative clustering) to determine whether stability advantages are hierarchical-structure-general or METIS-specific.

7.4 Enhanced Stability Metrics

Representative-Constituent Continuity: Measure what fraction of residents retain the same congressional representative (accounting for term limits, retirements). This captures political accountability dimension missing from geographic metrics.

Voter Surveys: Conduct surveys in states with recent redistricting to measure:

- Voter awareness of redistricting
- Confusion about district boundaries
- Perceived representation quality

Administrative Cost Modeling: Quantify actual costs of redistricting (updating voter rolls, precinct maps, election materials) and correlate with stability metrics.

7.5 Demographic Projection Integration

Proactive Stability Optimization: Incorporate demographic trend data to predict 2030 population distributions, then optimize 2020 districts for joint 2020-2030 stability:

1. Obtain demographic projections (Census Bureau, urban planning models)
2. Project 2030 tract populations
3. Run both 2020 and 2030 redistricting
4. Choose 2020 partition maximizing similarity to 2030 partition

Sensitivity Analysis: Test robustness to demographic projection errors—if projections are 10% wrong, does proactive optimization still help?

7.6 Generalization to Other Domains

School District Boundaries: Apply methods to school redistricting using enrollment data over time. Measure student reassignment rates and family disruption.

Service Territory Planning: Test hierarchical vs flat methods for hospital catchments, police precincts, fire department coverage areas using service demand shifts.

Dynamic Graph Partitioning Theory: Develop general theory of stability for time-varying graph partitioning, identifying conditions where hierarchical methods dominate flat methods.

7.7 Legal and Policy Research

Communities of Interest Doctrine: Collaborate with legal scholars to formalize "communities of interest" preservation as a legal criterion, potentially incorporating temporal stability metrics.

State Redistricting Standards: Survey state constitutional requirements for redistricting stability. Some states mandate "minimal change" from previous maps—our metrics could operationalize this requirement.

Public Perception Studies: Examine whether recursive bisection's interpretable tree structure improves public trust in algorithmic redistricting.

7.8 Immediate Next Steps

For authors wishing to build on this work immediately:

1. **Validate Results:** Reproduce our findings on the 5 southern states using our publicly available code and census data.

2. **Add 2000 Data:** Extend to three-cycle analysis (2000-2010-2020) for Alabama and Georgia (most data-complete states).

3. **Test Hybrid Method:** Implement hierarchical-initialized n-way and measure whether it achieves intermediate performance-stability balance.

4. **Expand to 10 States:** Add 5 more states with different geographic characteristics (e.g., Illinois, Pennsylvania, Ohio, Virginia, North Carolina) to test regional generalization.

By pursuing these directions, future work can build a comprehensive understanding of temporal dynamics in algorithmic redistricting, ultimately informing policy decisions balancing current optimization with long-term stability.

8 Conclusion

Congressional redistricting occurs every decade, yet research has focused almost exclusively on single-census optimization. This temporal myopia ignores a critical question: as demographics shift over ten years, how stable do district boundaries remain? We address this gap through the first systematic comparison of graph partitioning methods across census cycles.

Our analysis of five southern states over 2010-2020 reveals a fundamental finding: **hierarchical recursive bisection provides substantially greater temporal stability than flat n-way partitioning.** Recursive methods maintain 80% tract retention versus 70% for n-way (Cohen's $d = 1.2$, $p < 0.001$), reduce population disruption by 10 percentage points, and preserve 94% of top-level geographic splits. Communities of interest experience 14 percentage points less disruption.

This stability advantage stems from hierarchical structure preservation. Recursive bisection's binary tree creates nested regions reflecting fundamental geography (urban-rural divides, historical regions, natural boundaries). These top-level splits persist across decades because they encode geographic constants. When demographics shift, adjustments occur within regions without cascading across all boundaries. By contrast, n-way's flat structure re-optimizes all districts simultaneously, causing global disruption.

The implications are both practical and theoretical. **Practically**, jurisdictions face a performance-stability tradeoff: n-way offers +4.3 percentage points better 2020 minority concentration but -10 to -13 points worse 2020-2030 stability. For comfortable VRA margins, recursive bisection's stability benefits outweigh marginal performance costs. For borderline cases, n-way's immediate gains may dominate. Hybrid approaches—initializing n-way with recursive structure—could balance both objectives.

Theoretically, our findings generalize beyond redistricting. Any domain requiring periodic re-partitioning (school districts, service territories, sales regions) faces similar temporal dynamics. When re-partitioning frequency is low relative to timescale, stability gains value. Hierarchical methods provide continuity through structural preservation, while flat methods sacrifice disruption for per-instance optimization. This constitutes a fundamental tradeoff in dynamic partitioning problems.

Our work generates testable predictions for 2030 redistricting: states using recursive methods in 2020 should experience less 2030 disruption, and preserving 2020 tree structures should improve stability. When 2030 census data arrives, researchers can validate these predictions, testing our mechanistic explanation of hierarchical structure advantages.

Future research should extend temporally (adding 2000 data for three-cycle analysis), geographically (all 50 states), and algorithmically (hybrid methods, stability-aware optimization). Enhanced metrics—representative-constituent continuity, voter awareness surveys, administrative cost models—would capture socio-political dimensions beyond geographic stability. Integrating demographic projections could enable proactive stability optimization, jointly planning 2020 and predicted 2030 boundaries.

In sum: redistricting method choice should consider not only current optimization but also future continuity. Recursive bisection's hierarchical structure provides substantial temporal stability advantages—advantages likely to grow in importance as redistricting reform efforts emphasize community preservation and administrative efficiency alongside traditional criteria. When communities of interest matter, when administrative disruption imposes costs, and when long-term planning horizons extend beyond single census cycles, hierarchical methods merit serious consideration.

The question is no longer just "which method produces better districts today?" but "which method produces districts that remain coherent tomorrow?" Our findings suggest that for temporal stability, hierarchical structure makes all the difference.

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